

SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
    char input[200];
    int i;

    printf("Enter a string with escape characters: ");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");
    for (i = 0; input[i] != '\0'; i++) {
        if (input[i] == '\\\\' && input[i + 1] != '\0') {
            i++;
            switch (input[i]) {
                case 'n': printf("\n"); break;
                case 't': printf("\t"); break;
                case ' ': printf(" "); break;
                case '\\\\': printf("\\\\"); break;
                case '$': printf("$"); break;
                case '!': printf("!"); break;
                default: printf("%c", input[i]);
            }
        } else {
            printf("%c", input[i]);
        }
    }
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 6.1.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 6.1.c -o 6.1
ashishreddy@Ashishs-MacBook-Air ~ % ./6.1
Enter a string with escape characters: Hello\nWorld

Processed string:
Hello
World
```

Task 2:

Code:

```
#include <stdio.h>
```

```
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();
    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("Child process created.\n");
        printf("Child PID: %d\n", getpid());
        char *args[] = {"ls", "-l", NULL};
        execvp(args[0], args);
        // This runs only if execvp fails
        perror("execvp failed");
        exit(1);
    } else {
        // Parent process
        printf("Parent process.\n");
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child process finished.\n");
    }
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 6.2.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 6.2.c -o 6.2
ashishreddy@Ashishs-MacBook-Air ~ % ./6.2
Parent process.
Parent PID: 1124
Child process created.
Child PID: 1125
total 992
-rw-r--r--  1 ashishreddy ashishreddy    98 Jul 31 09:35 1.c
-rw-r--r--  1 ashishreddy ashishreddy    94 Jul 31 09:43 2.c
-rw-r--r--  1 ashishreddy ashishreddy   163 Jul 31 09:52 3.c
-rw-r--r--  1 ashishreddy ashishreddy   345 Jul 31 09:58 4.c
-rwxr-xr-x  1 ashishreddy ashishreddy 16176 Sep 21 14:31 5.1
-rw-r--r--  1 ashishreddy ashishreddy   385 Sep 21 14:31 5.1.c
-rwxr-xr-x  1 ashishreddy ashishreddy 16216 Sep 21 14:34 5.2
-rw-r--r--  1 ashishreddy ashishreddy   476 Sep 21 14:34 5.2.c
-rw-r--r--  1 ashishreddy ashishreddy   262 Aug 11 03:17 5.c
-rwxr-xr-x  1 ashishreddy ashishreddy 16032 Aug 11 03:21 6
-rwxr-xr-x  1 ashishreddy ashishreddy 16176 Sep 21 14:39 6.1
-rw-r--r--  1 ashishreddy ashishreddy  1042 Sep 21 14:39 6.1.c
-rwxr-xr-x  1 ashishreddy ashishreddy 16296 Sep 21 14:40 6.2
-rw-r--r--  1 ashishreddy ashishreddy   720 Sep 21 14:40 6.2.c
```